NEGROS ORIENTAL STATE UNIVERSITY

ITS 200 – Database Management System
ITS 201 – System Integration and Architecture

FINAL PROJECT DOCUMENTATION

Border Entry Payment and Personnel Tracking System

Group 3
BSINT 2nd Year – 1st Semester

Submitted by:
Dave Cordova
Kenn Caldoza
Kyfer Cainoy
Devine Cabanag
Albert Egay
Michael Fonollera

Submitted to:
Christian Franc M. Carvajal
Instructor

TABLE OF CONTENTS

1. Introduction and Background....3
2. Problem Statement....3
3. General and Specific Objectives....3
4. Scope and Limitations....3
5. System Requirements....3
6. Technologies Used....3
7. System Architecture....3
8. Database Design and ERD....3
9. System Implementation and Modules....3
    9.1 Travelers, Vehicles, Entry Records (Devine)....3
    9.2 Checkpoints, Lanes, Vehicle Types (Kyfer)....3
    9.3 Fee Rules, Penalties, Payments (Kenn)....3
    9.4 Personnel, Shifts, Checkpoint Assignments (Dave)....3
    9.5 Users and Roles (Albert)....4
    9.6 Violations, Alerts, Audit Logs (Michael)....4
10. CRUD Operations....4
11. SQL Queries and JOINs....4
12. Testing Procedures and Results....4
13. GitHub Collaboration....4
14. Conclusion and Recommendations....4
15. References....4
16. Appendices....4

1. Introduction and Background

[Explain what the system is, why it is needed, and who will use it.]

2. Problem Statement

[Describe the problem the system solves.]

3. General and Specific Objectives

[One general objective and 4 to 6 specific objectives.]

4. Scope and Limitations

[What the system covers and what it does not do.]

5. System Requirements

[Hardware and software needed to run the system.]

6. Technologies Used

[PHP, Node.js and Express, MySQL, HTML/CSS/JavaScript, Git and GitHub, Postman. Say how each one is used.]

7. System Architecture

[Insert System_Architecture.png (by Albert) and explain how the browser, PHP, Node.js API and MySQL connect.]

8. Database Design and ERD

[Insert ERD.png (by Kenn) and explain the tables and their relationships.]

9. System Implementation and Modules

[For each module write: purpose, main features, CRUD operations, tables used, important SQL queries, testing done, and limitations.]

9.1 Travelers, Vehicles, Entry Records (Devine)

[Write here.]

9.2 Checkpoints, Lanes, Vehicle Types (Kyfer)

[Write here.]

9.3 Fee Rules, Penalties, Payments (Kenn)

[Write here.]

9.4 Personnel, Shifts, Checkpoint Assignments (Dave)

[Write here.]

9.5 Users and Roles (Albert)

[Write here.]

9.6 Violations, Alerts, Audit Logs (Michael)

[Write here.]

10. CRUD Operations

[Summarize which Create, Read, Update and Delete functions work in each module.]

11. SQL Queries and JOINs

[Explain the six JOIN reports in the reports folder, one per member, with a screenshot of each result.]

12. Testing Procedures and Results

[List the tests that were actually done, the results, and any failures.]

13. GitHub Collaboration

[How the group used branches, pull requests and reviews.]

14. Conclusion and Recommendations

[What was achieved and what can be improved.]

15. References

[Websites and documentation used.]

16. Appendices

[Screenshots of the dashboard, modules, database, JOIN queries and API tests.]